



CLOUDPANDITH ONLINE TRAININGS



Master Data Engineering with Expert-Led Live Training

◇ Azure Data Factory

Complete ETL/ELT

◇ Azure Databricks

Big Data & PySpark

◇ Microsoft Fabric

End-to-End Analytics

📞 ENROLL: **8904424822** | 🌐 www.cloudpandith.com

✓ Live Sessions ✓ Real Projects ✓ Expert Trainers ✓ Interview Prep ✓ Lifetime Access

- ◆ Live Classes
- ◆ Real-Time Projects
- ◆ Interview Preparation
- ◆ Lifetime Access

Index

1. Delta Lake Architecture (Quick Recap)
2. Problem with Traditional DELETE & UPDATE
 - Full Parquet File Rewrite
 - Performance Issues
3. What is Deletion Vector (DV)? & Why Deletion Vector is Required
4. How Deletion Vector Works
 - Logical Deletion
 - Bitmap Concept
5. DELETE Operation using DV
6. UPDATE Operation using DV
7. DV Internal Working
 - DV Files
 - Row Tracking
8. Transaction Log Importance
 - JSON Files (Mandatory)
 - CRC Files (Optional)
9. File Handling Scenarios
 - Deleting Parquet Files
 - Deleting Log Files
 - Impact Analysis
11. ◆ Performance Benefits of DV
12. 🎯 Interview Summary

📞 8904424822 | 🌐 www.cloudpandith.com | ✉ cloudpandith@gmail.com

1. Delta Lake Architecture with Deletion Vector (DV)

Delta Lake is built on top of Data Lake storage and uses:

- 📁 Parquet files → Store actual data
 - 📄 Transaction logs (`_delta_log`) → Store metadata & history
 - 👉 Delta = Data (Parquet) + Logs (JSON/CRC)
-

2. ◇ Problem Without Deletion Vector

✗ Traditional Behavior (Before DV)

When you perform:

DELETE

UPDATE

- 👉 Delta rewrites the entire Parquet file

Example:

p1.parquet → 100 records

Delete 1 record

- ✗ System rewrites:

New file with 99 records

- 👉 Problems:

High compute cost
Slow performance
More file creation

3. ◇ What is Deletion Vector (DV)?

- 👉 Definition:

Deletion Vector is a feature in Delta Lake that allows:

- ✓ Logical deletion of rows
- ✓ Without rewriting the entire Parquet file

4. ♦ How Deletion Vector Works

☒ New Approach (With DV)

Instead of rewriting:

Keep Parquet file unchanged

Maintain a bitmap (DV file) to track deleted rows

Example:

p1.parquet → 100 records

Delete 1 record

👉 What happens:

p1.parquet → remains same

DV file → marks 1 row as deleted

✓ Faster

✓ Efficient

✓ No file rewrite

5. ♦ DV Internals

DV stores row-level deletion info

Implemented as bitmap structure

Stored alongside Delta logs

Representation:

p1.parquet → Original data (100 rows)

DV file → [0,0,1,0,0...] (1 = deleted row)

6. ♦ DELETE Operation with DV

Step 1: 100 records inserted → p1.parquet

Step 2: DELETE applied

Result:

- Parquet file → unchanged

- DV → marks deleted rows

👉 No rewrite happens

7. ✧ UPDATE Operation with DV

- 👉 UPDATE internally works as:
 - Old rows → marked as deleted (using DV)
 - New updated rows → written as new records
 - Old rows → DV marked
 - New rows → appended
- ✓ Efficient update handling

8. ✧ Transaction Log Importance

- 👉 Delta Lake cannot work without transaction logs
- Mandatory Files:
 - JSON files → actual metadata
 - _delta_log → version control
- 👉 If JSON logs are deleted:
 - ✗ Table becomes unreadable

9. ✧ Parquet vs Logs Behavior

Action	Result
Delete Parquet files manually	✗ Data loss (cannot recover)
Delete JSON logs	✗ Table unreadable
Delete CRC files	✓ No impact

- 👉 CRC files are optional (used for validation only)

10. Key Advantages of Deletion Vector

- ✓ No full file rewrite
 - ✓ Faster DELETE & UPDATE
 - ✓ Reduced compute cost
 - ✓ Better performance for large datasets
-

11. Interview Summary

👉 "Deletion Vector in Delta Lake allows logical deletion of rows using bitmap tracking without rewriting Parquet files, improving performance and reducing cost."